

NARENDRANATH EDARA

+1 (573) 466-6656 | edara.narendranath@gmail.com | linkedin.com/in/narendranathe | github.com/narendranathe | narendranathe.github.io

Data engineer building correct, recoverable pipelines for payroll-critical systems. Reengineered SQL Server change data capture from 30-minute full reloads to idempotent incremental loads under 8 minutes, with 67 percent less compute and clean reruns after failure. Owns change-controlled releases and incident recovery; earlier marketplace roles add direct ownership of unit economics and growth.

TECHNICAL SKILLS

Programming & Data: Python, SQL, T-SQL, SQL Server, stored procedures, change data capture (CDC), idempotent incremental loads, MERGE upserts

Reliability & Delivery: SQL Agent monitoring, SSIS execution history, Always On Availability Groups, recovery runbooks, Azure DevOps CI/CD, change control

Azure & Analytics: Azure Kubernetes Service, Microsoft Fabric semantic models, Power BI, Azure AI Anomaly Detector, dimensional modeling

Distributed Data: Apache Spark, Delta Lake, Delta tables, lakehouse architecture, Kafka, FastAPI, MLflow, Airflow, Docker

EXPERIENCE

Data Engineer | *ExponentHR*

July 2024 - Present | Dallas, TX

Payroll platform for client tenants: owns CDC ingestion and recovery across 400 SQL Server databases, about 70 TB on 10 servers.

- Cut payroll data refresh from 30 minutes to **under 8 minutes** by replacing full-table reloads with idempotent change data capture (CDC) MERGE upserts; reduced compute 67 percent while keeping failed loads safe to rerun, now the platform standard.
- Freed **5,000 engineering hours** a year by making Contained Always On Availability Group database copy and restore safe to retry; fixed orphaned CDC jobs, active replication races, silent parameter defaults, and stuck restores.
- Protected payroll balances by tracing a paid-time-off accrual defect to a payroll-period boundary in SQL, then shipping the corrected calculation to every client tenant through formal **change-request review** instead of a direct patch.
- Shipped SECURE 2.0 retirement compliance by its legislated deadline alongside a Clock Out Type enhancement.
- Compressed releases from 3 months to **14 days** by owning Azure DevOps release automation (CI/CD), automating reporting steps, and aligning dev, test, IT, and business on Scrum priorities; removed about 11 weeks of idle handoff time per release.
- Raised payroll-platform uptime from **98 to 99.9 percent** by instrumenting SQL Agent jobs for missed, stuck, and silent failures, then shipping self-healing recovery before expired CDC logs could make failures unrecoverable.
- Made silent data drift alertable with SQL Server package run history, control-table checks, and a named response owner.
- Cut post-failover CDC capture-job recovery to **under one hour** by documenting Always On Availability Group runbooks and closing a silent job failure; made a specialist-only restore repeatable for on-call engineers.
- Cut report queries from 12 seconds to under 4 and client support tickets 40 percent by migrating SQL Server reporting to **Microsoft Fabric** semantic models and tuning indexes and stored procedures on analytical paths.
- Automated delivery reporting with Python workflows over Git and Azure DevOps APIs, giving the team real-time release visibility and saving 15+ engineering hours per sprint, or about 390 hours annually.

Data Engineer | *Missouri S&T*

Aug. 2023 - July 2024 | Rolla, MO

University research infrastructure for anomaly detection; owned Azure monitoring, incident signals, and cloud cost controls.

- Caught a production memory leak four hours before an outage by engineering Azure AI Anomaly Detector pipelines that selected algorithms per time-series profile; reached more than 95 percent accuracy across research workloads.
- Filtered about 250 non-actionable priority-3 alerts each week with tunable per-service thresholds, moving signal-to-noise from 1:5 to 1:1.2 and reducing on-call alert blindness that had hidden actionable incidents.
- Cut Azure spend \$3,200 per month by moving anomaly detection from 20 static VMs to **Azure Kubernetes Service** with horizontal pod autoscaling; lifted CPU utilization from 12 to 64 percent while running 4 to 8 nodes.

Product Intern | *C2FO*

June 2023 - Aug. 2023 | Kansas City, KS

B2B fintech working-capital platform: analyzed transaction behavior and defined preferred-offer requirements for product teams.

- Used SQL to analyze transaction and user behavior on a B2B working-capital platform, guiding product decisions.
- Cut resource-allocation time 50 percent by authoring preferred-offers product requirement documents from stakeholder interviews.

Business Intelligence Analyst | *Udaan*

Sept. 2020 - Mar. 2021 | Nashik, India

New-city B2B staples marketplace: owned gross merchandise value, demand planning, and procurement workflows for the launch.

- Grew Nashik city gross merchandise value to INR 5 crore a month during the new-city Staples launch, matching the incumbent; reached INR 3.9 crore in December at **1.3 times** month-over-month growth while owning procurement and category planning.
- Sustained 99.3 percent order fulfillment under just-in-time and warehouse models by applying statistical demand analysis to capacity planning for the new-city Staples category and its changing buyer demand.
- Kept delivery cycles running through application downtime by devising and executing a business continuity plan for Nashik.
- Raised ROI 7 percent through demand forecasts and inventory models built in Power BI and Excel macros for Nashik Staples.
- Hired and managed a 10+ person sourcing team for the Nashik Staples launch, covering procurement as the city business scaled.

Business Analyst | *Zomato*

Mar. 2018 - Sept. 2020 | Hyderabad, India

Consumer marketplace across a 300-restaurant partner portfolio; owned pricing intelligence, search analysis, and unit economics.

- Turned **contribution per order** from negative 18 to positive 2 rupees across 300 restaurants by rebalancing discount sharing against advertising revenue, moving the portfolio from loss to positive unit economics.
- Fed pricing with current competitor signals on price, delivery time, and coverage by building real-time analytics with an ingestion and refresh cadence; contributed to a 9 percent market-share gain in contested metros.
- Cut operations support-desk volume 80 percent by indexing more than 100,000 internal documents in Elasticsearch for self-service search, so teams could resolve common questions without waiting on support.
- Delivered 14 percent compounded quarterly growth across 300 restaurants by lifting campaign coverage to 70 percent.

PROJECTS

repo-context-hooks | *Python, GitHub Actions, Sigstore, CodeQL*

PyPI

- Published a zero-dependency Python package that snapshots repository context before and after commits, letting LLM agents resume work across sessions without imposing runtime dependencies on downstream projects.
- Hardened its release path with Sigstore-signed OpenID Connect publishing, CodeQL on each pull request, and Dependabot for GitHub Actions; validated the public package with **330+ passing tests** at v1.0 to protect the release path.

Real-Time Fraud Detection Platform | *Kafka, FastAPI, LightGBM, MLflow*

2026

- Built Kafka-to-FastAPI fraud scoring on 100,000 **synthetic transactions** with MLflow-tracked LightGBM, exactly-once processing, and Prometheus/Grafana monitoring; sustained 100+ transactions per second at 1.12 ms 99th-percentile latency.

FinTune | *Mistral-7B, QLoRA, FastAPI, PyTorch*

2026

- Built financial-sentiment serving with QLoRA-fine-tuned Mistral-7B and FastAPI, adding pre-inference personal-data redaction and drift monitoring; 35+ tests cover data, model, guardrails, serving, and recovery.
- Added a three-state circuit breaker that reloads the model, reduces batch size under memory pressure, or falls back to a quantized model under sustained failure, so inference can recover from faults instead of remaining unavailable.

AutoApply AI | *Chrome MV3, FastAPI, pgvector, React*

2026

- Built a privately deployed workflow with pgvector-grounded answer retrieval and 11 applicant-tracking-system (ATS) adapters; shipped 40+ backend endpoints and **355 tests**, with public launch held for security fixes.

Portfolio Risk Analytics (in progress) | *Kafka, Spark Structured Streaming, FastAPI*

2026

- Built Kafka market-tick ingestion with Spark Structured Streaming in 5-second tumbling windows and a FastAPI prototype for historical-simulation VaR at 95 and 99 percent confidence; end-to-end handoff remains in progress.

EDUCATION & CERTIFICATIONS

Missouri S&T - MS Information Science and Technology, GPA 4.0/4.0

Jan. 2022 - Dec. 2023

GVP College of Engineering - B.Tech Mechanical Engineering

2017

DP-700 Microsoft Certified Fabric Data Engineer Associate | CSPO | Publication DOI 10.1080/10495142.2025.2525123